

Quantum Symmetry via K-theory

Bowen Yang, joint work with Mattie Ji

Harvard CMSA
Department of Mathematics, Harvard University

Algebraic Structures in Topology 2026

Joint work with Mattie Ji

Based on:

- *Quantum Cellular Automata: The Group, the Space, and the Spectrum*
Mattie Ji and Bowen Yang, arXiv:2602.16572.
- *K-Theoretic Obstructions to Linearizing QCA Representations*
Mattie Ji and Bowen Yang, arXiv:2606.19657.

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

- 1 Gapped Phases and QCA
- 2 Symmetries and “Anomalies”
- 3 Obstruction Theory

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

- 1 Gapped Phases and QCA
- 2 Symmetries and “Anomalies”
- 3 Obstruction Theory

Quantum Spin Systems and Gapped Phases

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Quantum spin system. A quantum spin system consists of local degrees of freedom placed on a lattice X , for example $X = \mathbb{Z}^d$.

- Observables form a C^* -algebra $\mathcal{A}(X)$.
- A state

$$\omega : \mathcal{A}(X) \rightarrow \mathbb{C}$$

assigns expectation values to observables.

Quantum Spin Systems and Gapped Phases

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Quantum spin system. A quantum spin system consists of local degrees of freedom placed on a lattice X , for example $X = \mathbb{Z}^d$.

- Observables form a C^* -algebra $\mathcal{A}(X)$.
- A state

$$\omega : \mathcal{A}(X) \rightarrow \mathbb{C}$$

assigns expectation values to observables.

Gapped state. A state is gapped if it has stable low-energy behavior, represented physically by an energy gap above the ground-state sector.

Quantum Spin Systems and Gapped Phases

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

Quantum spin system. A quantum spin system consists of local degrees of freedom placed on a lattice X , for example $X = \mathbb{Z}^d$.

- Observables form a C^* -algebra $\mathcal{A}(X)$.
- A state

$$\omega : \mathcal{A}(X) \rightarrow \mathbb{C}$$

assigns expectation values to observables.

Gapped state. A state is gapped if it has stable low-energy behavior, represented physically by an energy gap above the ground-state sector.

Phase. A phase is an equivalence class of gapped states under continuous paths that preserve the gap. Phases form a monoid with $\mathcal{I}(\mathbb{Z}^d)$ its group of units.

Quantum Spin Systems and Gapped Phases

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Quantum spin system. A quantum spin system consists of local degrees of freedom placed on a lattice X , for example $X = \mathbb{Z}^d$.

- Observables form a C^* -algebra $\mathcal{A}(X)$.
- A state

$$\omega : \mathcal{A}(X) \rightarrow \mathbb{C}$$

assigns expectation values to observables.

Gapped state. A state is gapped if it has stable low-energy behavior, represented physically by an energy gap above the ground-state sector.

Phase. A phase is an equivalence class of gapped states under continuous paths that preserve the gap. Phases form a monoid with $\mathcal{I}(\mathbb{Z}^d)$ its group of units.

Central Expectation

Gapped phases of quantum matter should be described by
topological quantum field theories.

Warning!

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Central Expectation

Gapped phases of quantum matter should be described by
topological quantum field theories.

This is not a **conjecture**. There are known exceptions under the umbrella term
"facton phases".

Warning!

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Central Expectation

Gapped phases of quantum matter should be described by
topological quantum field theories.

This is not a **conjecture**. There are known exceptions under the umbrella term
"fracton phases".

Beaudry: From Stabilizer Codes to Fracton Phases of Matter



 Jun 15, 2026, 9:00 AM

 1h 30m

 Institut Henri Poincaré

Description

Fracton orders are quantum systems characterized by the presence of excitations with restricted mobility. Many examples are described by stabilizer codes, that is, lattice Hamiltonians constructed from commuting generalized Pauli matrices. In this lecture series, we will explore the algebraic framework for fracton phases and mutual statistics that emerges from the study of stabilizer codes.

The course will be mostly based on work of Haah and Ruba-Yang, as well as joint work with Hermele, Shirley, Wickenden and Yang.

Figure: Available on YouTube

Invertible Phases

Nevertheless, the invertible phases are expected to correspond to invertible TQFTs.

Theorem/Conjecture (Kitaev, Freed–Hopkins, Kapustin, Kubota...)

Invertible phases assemble into an Ω -spectrum

$$\mathbb{IP} = \{\mathrm{IP}(\mathbb{Z}^d)\}_{d \geq 0}, \quad \pi_0(\mathrm{IP}(\mathbb{Z}^d)) \cong \mathcal{I}(\mathbb{Z}^d).$$

This spectrum is expected to agree with the Freed–Hopkins spectrum classifying invertible TQFTs.

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

Invertible Phases

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Nevertheless, the invertible phases are expected to correspond to invertible TQFTs.

Theorem/Conjecture (Kitaev, Freed–Hopkins, Kapustin, Kubota...)

Invertible phases assemble into an Ω -spectrum

$$\mathbb{IP} = \{\mathrm{IP}(\mathbb{Z}^d)\}_{d \geq 0}, \quad \pi_0(\mathrm{IP}(\mathbb{Z}^d)) \cong \mathcal{I}(\mathbb{Z}^d).$$

This spectrum is expected to agree with the Freed–Hopkins spectrum classifying invertible TQFTs.

Theorem 1.1. *There is a 1:1 correspondence*

$$(1.2) \quad \left\{ \begin{array}{l} \text{deformation classes of reflection positive} \\ \text{invertible } n\text{-dimensional extended topological} \\ \text{field theories with symmetry group } H_n \end{array} \right\} \cong [MTH, \Sigma^{n+1} I\mathbb{Z}(1)]_{\mathrm{tor}}.$$

Figure: Freed–Hopkins 2016

Quantum Cellular Automata (QCA)

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Definition: A QCA in dimension d is a strictly locality-preserving automorphism

$$\alpha : \mathcal{A}(\mathbb{Z}^d) \rightarrow \mathcal{A}(\mathbb{Z}^d).$$

Structure¹:

- QCA form a group $\mathcal{Q}^*(\mathbb{Z}^d)$ under composition
- Contains a normal subgroup $\mathcal{C}^*(\mathbb{Z}^d)$ of quantum circuits

$\mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d)$ is abelian (Freedman–Hastings).

- A QCA representation is a homomorphism

$$\varphi : G \longrightarrow \mathcal{Q}^*(\mathbb{Z}^d).$$

¹After adding arbitrary ancilla to stabilize

QCA action on States

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Given a state

$$\omega : \mathcal{A}(\mathbb{Z}^d) \rightarrow \mathbb{C},$$

a QCA α produces a new state

$$\omega \circ \alpha : \mathcal{A}(\mathbb{Z}^d) \rightarrow \mathcal{A}(\mathbb{Z}^d) \rightarrow \mathbb{C}.$$

This means a QCA representation defines a group action on the quantum system or a *quantum G-symmetry*.

Key fact:

- Circuits $\mathcal{C}^*(\mathbb{Z}^d)$ do not change phase

Hence we obtain a group homomorphism

$$\mathcal{Q}^*(\mathbb{Z}^d)/\mathcal{C}^*(\mathbb{Z}^d) \longrightarrow \mathcal{I}(\mathbb{Z}^d).$$

QCA probes invertible phases.

Theorem (Ji-Y.)

Using algebraic K -theory, for a nice enough metric space X , we construct an Ω -spectrum

$$\mathbf{QCA}(X) = \{\mathbf{QCA}(X \times \mathbb{Z}^d)\}_{d \geq 0}.$$

such that

$$\pi_0(\mathbf{QCA}(X \times \mathbb{Z}^d)) \cong \mathcal{Q}^*(X \times \mathbb{Z}^d)/\mathcal{C}^*(X \times \mathbb{Z}^d).$$

Theorem (Ji-Y.)

Using algebraic K -theory, for a nice enough metric space X , we construct an Ω -spectrum

$$\mathrm{QCA}(X) = \{\mathrm{QCA}(X \times \mathbb{Z}^d)\}_{d \geq 0}.$$

such that

$$\pi_0(\mathrm{QCA}(X \times \mathbb{Z}^d)) \cong \mathcal{Q}^*(X \times \mathbb{Z}^d)/\mathcal{C}^*(X \times \mathbb{Z}^d).$$

Theorem/Conjecture (Kitaev, Freed–Hopkins, Kapustin, Kubota,...)

Invertible phases assemble into an Ω -spectrum

$$\mathbb{IP} = \{\mathrm{IP}(\mathbb{Z}^d)\}_{d \geq 0}, \quad \pi_0(\mathrm{IP}(\mathbb{Z}^d)) \cong \mathcal{I}(\mathbb{Z}^d).$$

This spectrum is expected to agree with the Freed–Hopkins spectrum classifying invertible TQFTs.

The Elephant in the Room

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

How is QCA related to $\mathbb{I}\mathbb{P}$
and the Freed–Hopkins spectrum?

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

How is QCA related to $\mathbb{I}\mathbb{P}$
and the Freed–Hopkins spectrum?
We don't know yet

The Elephant in the Room

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

How is QCA related to $\mathbb{I}\mathbb{P}$
and the Freed–Hopkins spectrum?

We don't know yet

Nevertheless, the QCA spectrum has other important
applications.



Table of Contents

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

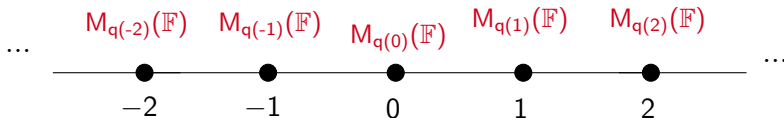
① Gapped Phases and QCA

② Symmetries and “Anomalies”

③ Obstruction Theory

Algebra of Observables

$$X = \mathbb{Z}$$



More precisely,

$$\mathcal{A}(X, q) = \operatorname{colim}_{\text{finite subsets } S \subseteq X} \underbrace{\bigotimes_{s \in S} M_{q(s)}(\mathbb{F})}_{=:\mathcal{A}(S, q)},$$

where for $S \subseteq T$ finite subsets,

$$\mathcal{A}(S, q) \rightarrow \mathcal{A}(T, q)$$

is given by $- \otimes \operatorname{id}_{\mathcal{A}(T-S, q)}$, tensoring with the identity operator on $T - S$.

Quantum Cellular Automata

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

A quantum cellular automaton α is an algebra automorphism

$$\alpha : \mathcal{A}(X, q) \rightarrow \mathcal{A}(X, q)$$

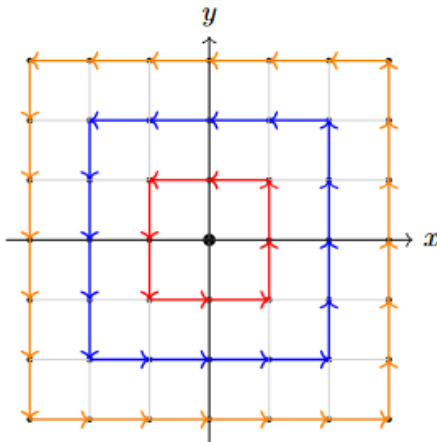
of finite spread, meaning that there exists some $\ell > 0$ such that for all $\gamma \in M_{q(x)}(\mathbb{F})$, $\alpha(\gamma) \in \mathcal{A}(D_\ell(x), q)$.

In other words, a QCA is an “automorphism that moves components away in a distance that is uniformly bounded”.

Definition:

The group of QCAs over $\mathcal{A}(X, q)$ is denoted $\mathcal{Q}(X, q)$.

Annular shift on $\mathbb{Z}^2$ (2D example)



Examples

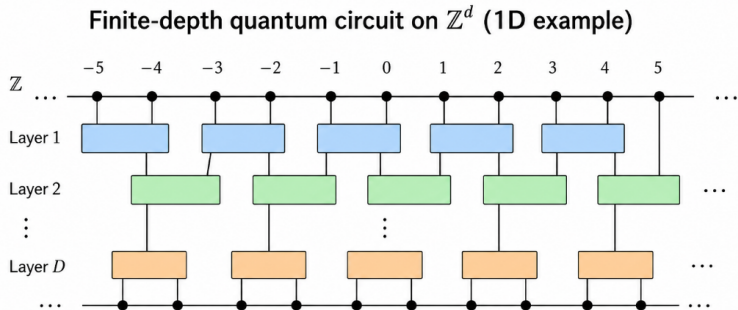
Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory



A special case: onsite QCA given by $\alpha \in \prod_{x \in X} PGL_{q_x}(\mathbb{F})$ or $\prod_{x \in X} PU(q_x)$.

Linearization of QCA representation

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Given a QCA representation $\varphi : G \rightarrow \mathcal{Q}(X, q)$. Does there exist a QCA $\beta \in \mathcal{Q}(X, q)$ such that the following diagram can be completed?

$$\begin{array}{ccc}
 & \prod_{x \in X} \mathrm{GL}_{q_x}(\mathbb{F}) & \\
 \nearrow & \downarrow & \\
 G & \prod_{x \in X} \mathrm{PGL}_{q_x}(\mathbb{F}) & \\
 \nearrow & \downarrow & \\
 G & \xrightarrow{\beta \varphi \beta^{-1}} & \mathcal{Q}(X, q)
 \end{array}$$

Figure: Linearization

Question (Linearization). Does there exist a lift $\tilde{\varphi}$ to the following diagram?

$$\begin{array}{ccc} & GL(V) & \\ \tilde{\varphi} \nearrow & \downarrow & \\ G & \xrightarrow{\varphi} & PGL(V). \end{array}$$

This question has a well-known answer in terms of an obstruction class in group cohomology.

Theorem 1.1. *Let $\varphi : G \rightarrow PGL(V)$ be a projective representation over a field $\mathbb{F}$. Choose lifts $U_g \in GL(V)$ of $\varphi(g)$. Then there are scalars $\alpha(g, h) \in \mathbb{F}^\times$ such that*

$$U_g U_h = \alpha(g, h) U_{gh}.$$

The function α is a 2-cocycle, and its cohomology class

$$[\alpha] \in H^2(G, \mathbb{F}^\times)$$

is independent of the choice of lifts. The projective representation φ is linearizable if and only if $[\alpha] = 0$.

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
“Anomalies”

Obstruction
Theory

- 1 Gapped Phases and QCA
- 2 Symmetries and “Anomalies”
- 3 Obstruction Theory

Stabilization

A QCA representation φ is *(trivially) stably linearizable* if $\varphi \otimes \psi_0 : G \rightarrow \mathcal{Q}(X, qr)$ admits a linearization where $\psi_0 : G \rightarrow \mathcal{Q}(X, r)$ is the trivial representation.

Theorem(Ji-Y.)

Let G be a finite or, more generally, rationally acyclic group. A QCA representation $\varphi : G \rightarrow \mathcal{Q}(X, q)$ defines a *cohomology class* $[\alpha] \in \text{QCA}(X)^1(BG)$ which acts as an obstruction to stable linearization.

Stable linearization and QCA spectrum

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Stabilization

A QCA representation φ is (*trivially*) *stably linearizable* if $\varphi \otimes \psi_0 : G \rightarrow \mathcal{Q}(X, qr)$ admits a linearization where $\psi_0 : G \rightarrow \mathcal{Q}(X, r)$ is the trivial representation.

Theorem(Ji-Y.)

Let G be a finite or, more generally, rationally acyclic group. A QCA representation $\varphi : G \rightarrow \mathcal{Q}(X, q)$ defines a *cohomology class* $[\alpha] \in \mathbb{QCA}(X)^1(BG)$ which acts as an obstruction to stable linearization.

Let's contrast it with the classical result.

The classical result

Let $\varphi : G \rightarrow PGL(V)$ be a projective representation over a field $\mathbb{F}$. The projective representation φ is linearizable only if the induced $[\alpha] \in H^2(G, \mathbb{F}^\times)$ vanishes.

Obstruction classes in ordinary cohomology

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Theorem(Ji-Y.)

When G is finite (or more generally, rationally acyclic) and $X = \mathbb{Z}^n$, there are successive obstruction classes in

$$H^{i+1}(BG; \mathcal{Q}^*(\mathbb{Z}^{n-i})/\mathcal{C}^*(\mathbb{Z}^{n-i})),$$

for $0 \leq i \leq n$, and in

$$H^{n+2}(BG; U(1)).$$

Obstruction classes in ordinary cohomology

Quantum
Symmetry via
K-theory

Bowen Yang,
joint work with
Mattie Ji

Gapped Phases
and QCA

Symmetries and
"Anomalies"

Obstruction
Theory

Theorem(Ji-Y.)

When G is finite (or more generally, rationally acyclic) and $X = \mathbb{Z}^n$, there are successive obstruction classes in

$$H^{i+1}(BG; \mathcal{Q}^*(\mathbb{Z}^{n-i})/C^*(\mathbb{Z}^{n-i})),$$

for $0 \leq i \leq n$, and in

$$H^{n+2}(BG; U(1)).$$

Proof (sketch): The obstruction class $[\alpha] \in \mathbb{QCA}(\mathbb{Z}^n)^1(BG)$ corresponds to a based map, up to homotopy,

$$BG \xrightarrow{\varphi^{\text{st}}} B\mathcal{Q}(\mathbb{Z}^n) \xrightarrow{g} \mathbb{QCA}(\mathbb{Z}^{n+1}).$$

Construct the Postnikov tower of the homotopy fiber of g . The above classes are obstructions to lifting φ^{st} up this tower.